

Customer Shopping Behavior Analysis

- Project Overview

- In this project, I performed EDA and Data analysis using python and SQL and created a Dashboard in Power bi. The purpose is to understand customer behavior to make data driven decisions by analyzing seasonal trends, demographics patterns and categories.

- Dataset Summary

- 3900 rows and 18 columns
- 2 derived columns
- 1 column is dropped
- Missing values are imputed in Review Rating Column

- EDA & Feature Engineering Using Python

- Import Libraries and Load the Dataset
- Analyze Shape and Dataset info
 - Using df.shape and df.info()
- Handling Missing values in Review Rating Column
 - Imputed the missing values by the median of review rating of each category

- Column name Standardization

- After Standardization

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
      'purchase_amount', 'location', 'size', 'color', 'season',  
      'review_rating', 'subscription_status', 'shipping_type',  
      'discount_applied', 'promo_code_used', 'previous_purchases',  
      'payment_method', 'frequency_of_purchases'],  
      dtype='object')
```

- Derived Age group column
 - Head rows

	age	age_group
0	55	Middle-aged
1	19	Young Adult
2	50	Middle-aged
3	21	Young Adult
4	45	Middle-aged

- Map purchase frequency days to respective days
- Remove unnecessary column (promo_code_used)
- Load Dataset into the Database (MySQL)

● Business Analysis using SQL

- Revenue By Gender

	gender	Total Revenue
▶	Male	157890
	Female	75191

- High Spending Discount Customers

	customer_id	purchase_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90

- Top 5 products with Highest Average Review Rating

	item_purchased	avg_review_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

- Standard Vs Express Shipping Avg Purchase Amount Comparison

	shipping_type	Avg_purchase_amount
▶	Express	60.48
	Standard	58.46

- Avg spend and total revenue b/w subscribed and non-subscribed customers

	subscription_status	Total Customers	avg_spend	total spend
▶	No	2847	59.87	170436
	Yes	1053	59.49	62645

- Products with highest Percentage of Purchase with discount applied

	item_purchased	Purchased %
▶	Pants	2.08
	Sweater	2.03
	Coat	2.03
	Hat	1.97
	Jewelry	1.95

- Customer Segment Comparison

	customer_segment	customer_count
▶	Loyal	3116
	New	83
	Returning	701

- Top 3 most purchased products within each category

	product_rank	category	item_purchased	purchased_count
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

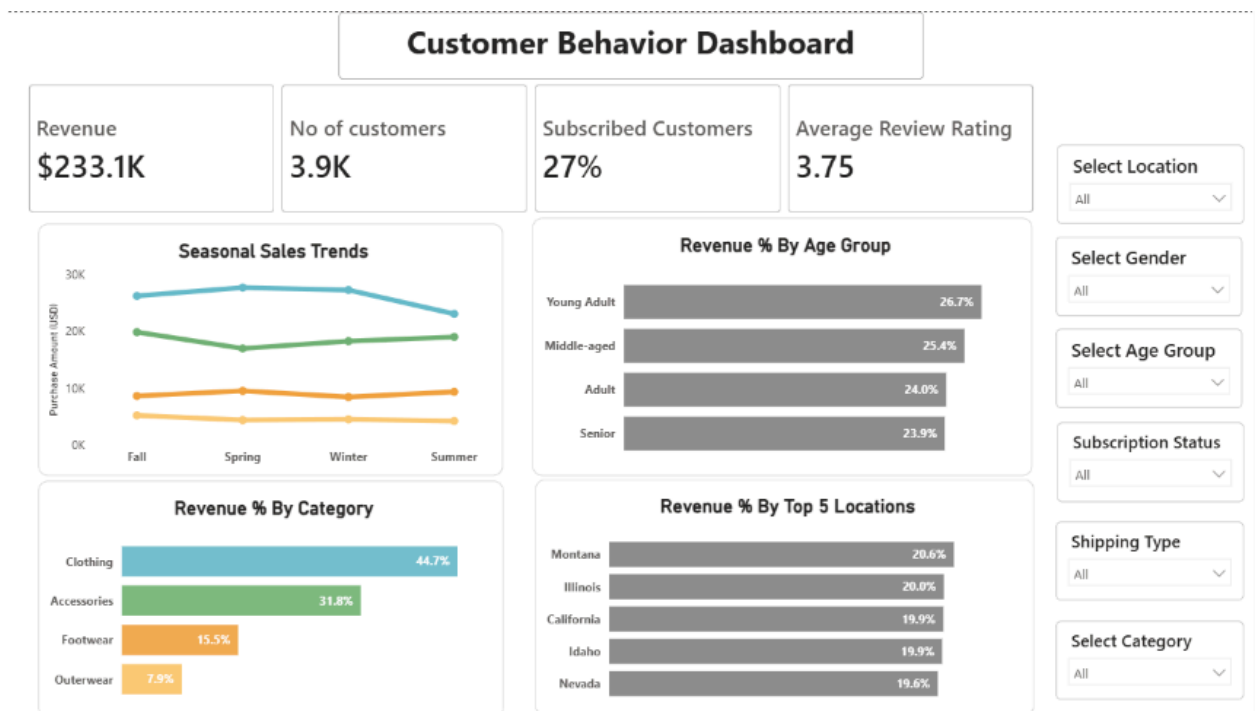
- Are customers who are repeat buyers (having more than 5 previous purchases) are more likely to subscribe?

	subscription_status	count(customer_id)
▶	Yes	958
	No	2518

- Revenue Contribution of each age group

	age_group	revenue %
▶	Middle-aged	25.40
	Young Adult	26.66
	Senior	23.92
	Adult	24.02

● Dashboard in Power Bi



● Business Recommendation

- We should Focus on Female products (As skirts have the least average rating) and offer them discounts
- We should make subscription offers more attractive.
- We should improve footwear and outwear products and marketing strategy